ON-DEVICE STATISTICAL EDGE PROCESSING

Sensor Sampling Layer

- Temperature
- Humidity
- CO2
- TVOC
- eCO2
- PM1.0
- PM2.5
- PM4.0
- PM10

Environmental sensors continuously generate raw telemetry data.

Local Statistical Computation

3 Minute Rolling Window
Calculated directly on MCU

Computed Features:

- Mean
- Standard Deviation
- Maximum Value

The microcontroller continuously performs local statistical aggregation before transmission.

Compact Binary Packet Generation

Fixed-order CBOR Binary Array



Packet contains:

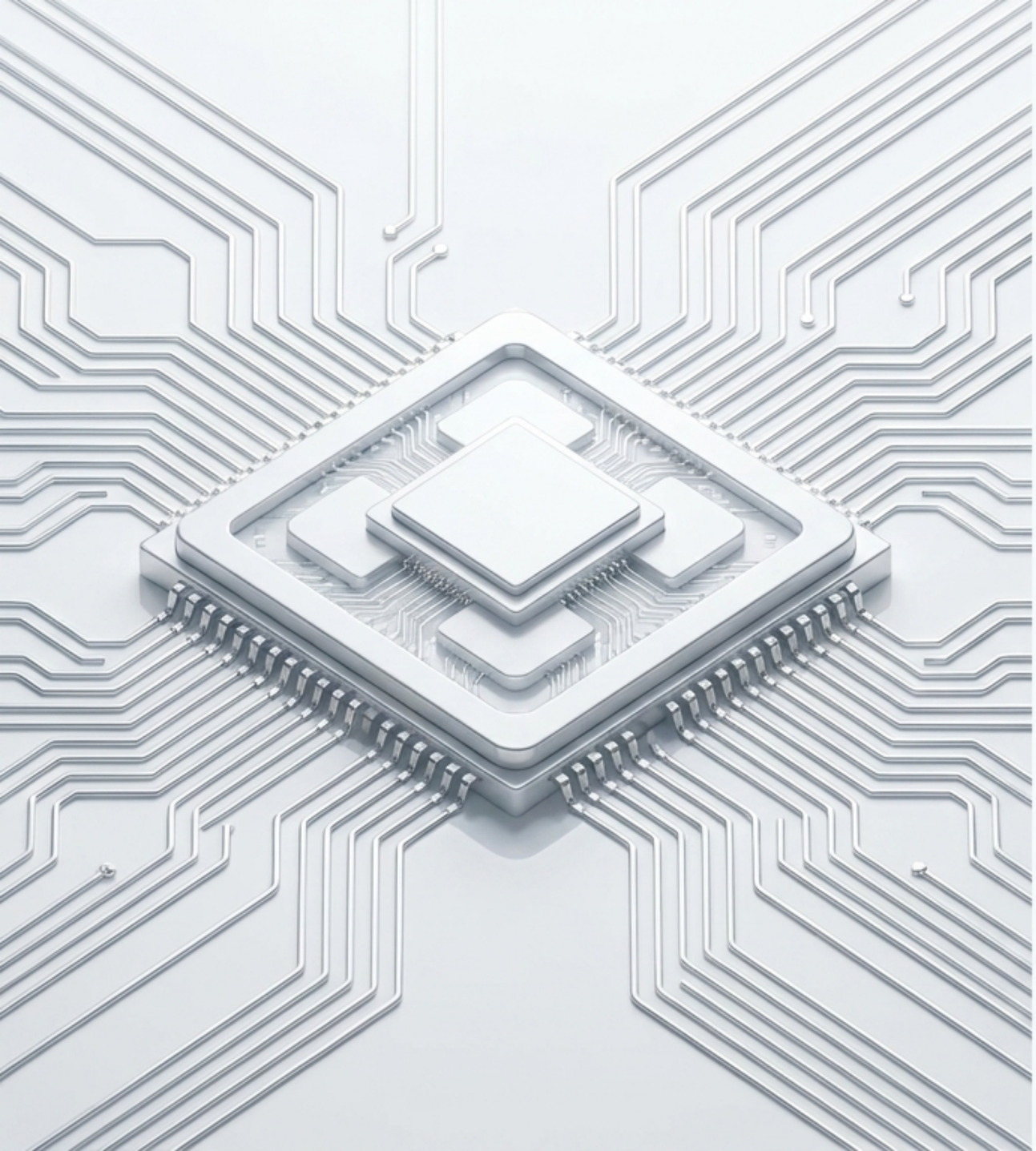
- Raw Sensor Values
- Statistical Features
- Hardware Diagnostics
- Mesh Metadata
- Sequence Number
- Boot ID
- Uptime Metadata

43 encoded values transmitted every 30 seconds

Data is serialized into a compact binary packet optimized for low-bandwidth mesh communication.

Sensors do not transmit raw telemetry blindly. Each sensor performs local edge computation and statistical aggregation before generating compact CBOR binary packets optimized for efficient OpenThread mesh transmission.

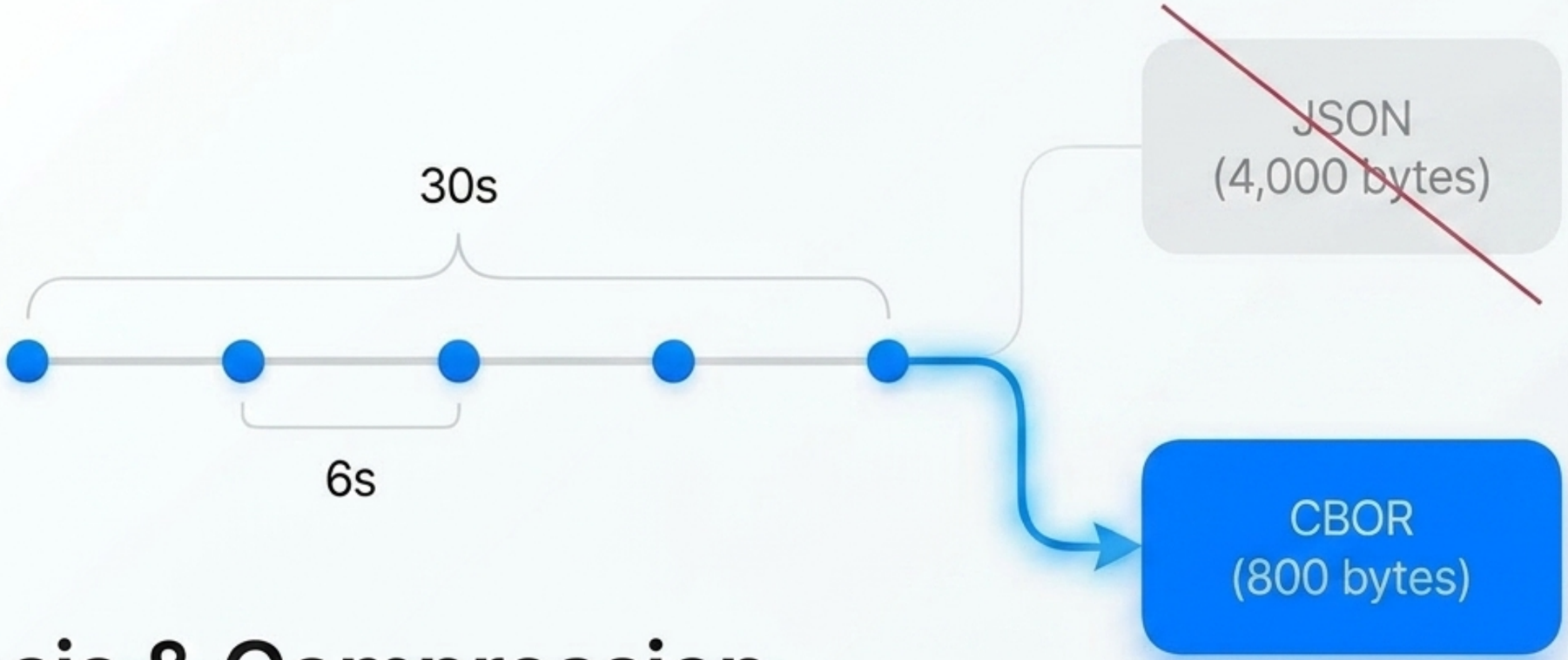
NotebookLM



The Hardware Edge

- **64 MHz ARM Cortex-M4:** High-efficiency edge processing via Nordic nRF52840 (running Zephyr RTOS).
- **1 MB Flash / 256 KB RAM:** Deep local data buffering for network outages.
- **IEEE 802.15.4:** Native 2.4 GHz radio support for OpenThread mesh networking.

NotebookLM



Data Genesis & Compression

- **Hardware Limit:** 5.5s mandatory refresh time between physical CO_2 sensor readings.
- **Batch Strategy:** 5 distinct environmental samples bundled every 30 seconds to minimize radio traffic.
- **CBOR Compression:** 80% payload size reduction, highly optimized for low-power wireless networks.

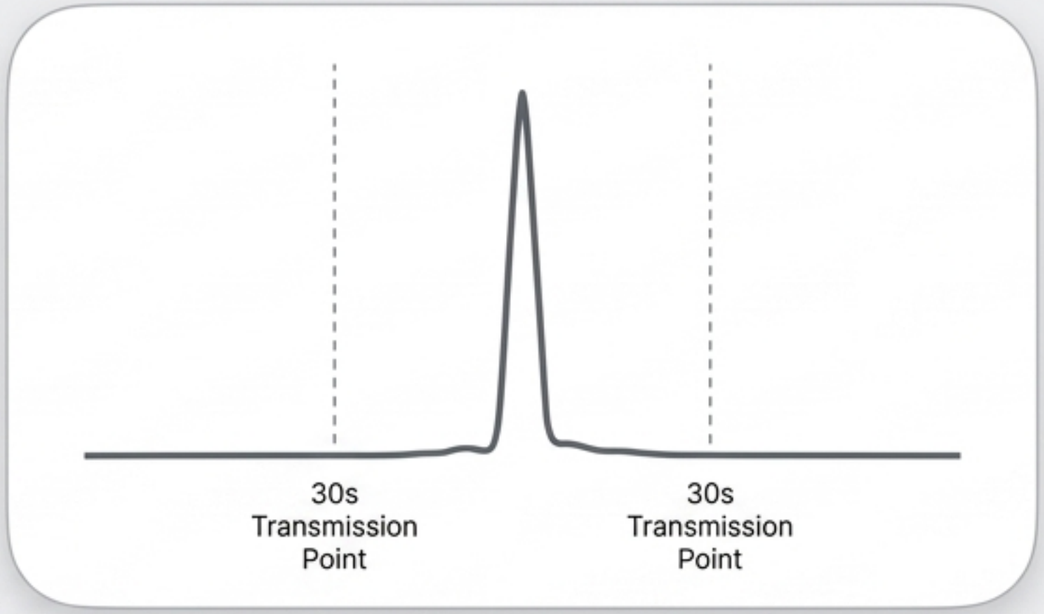
NotebookLM

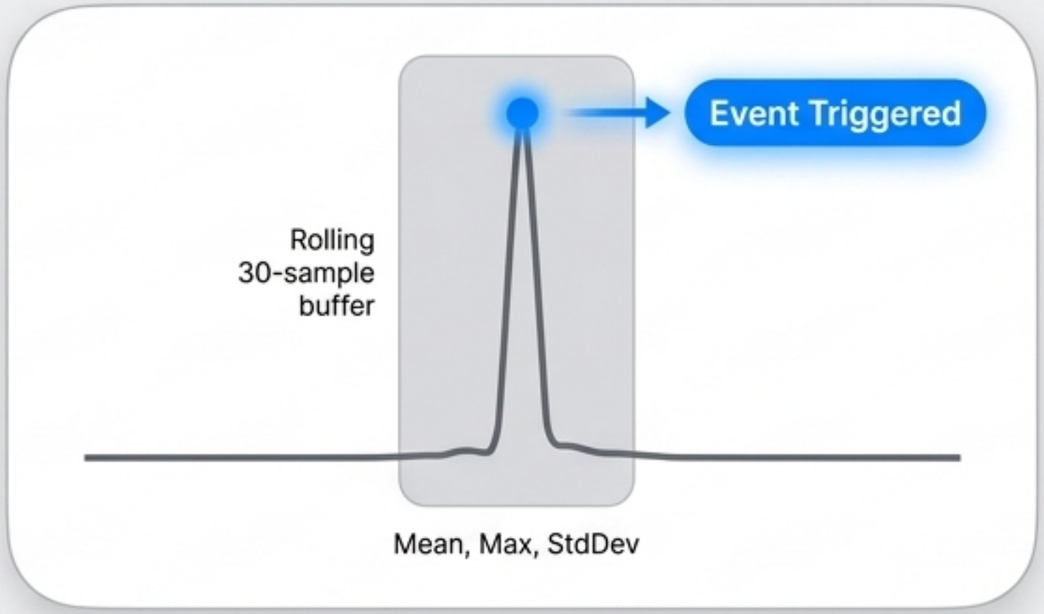
Edge Intelligence & Event Awareness

Continuous Buffer: Firmware maintains a rolling 3-minute memory window to capture transient spikes.

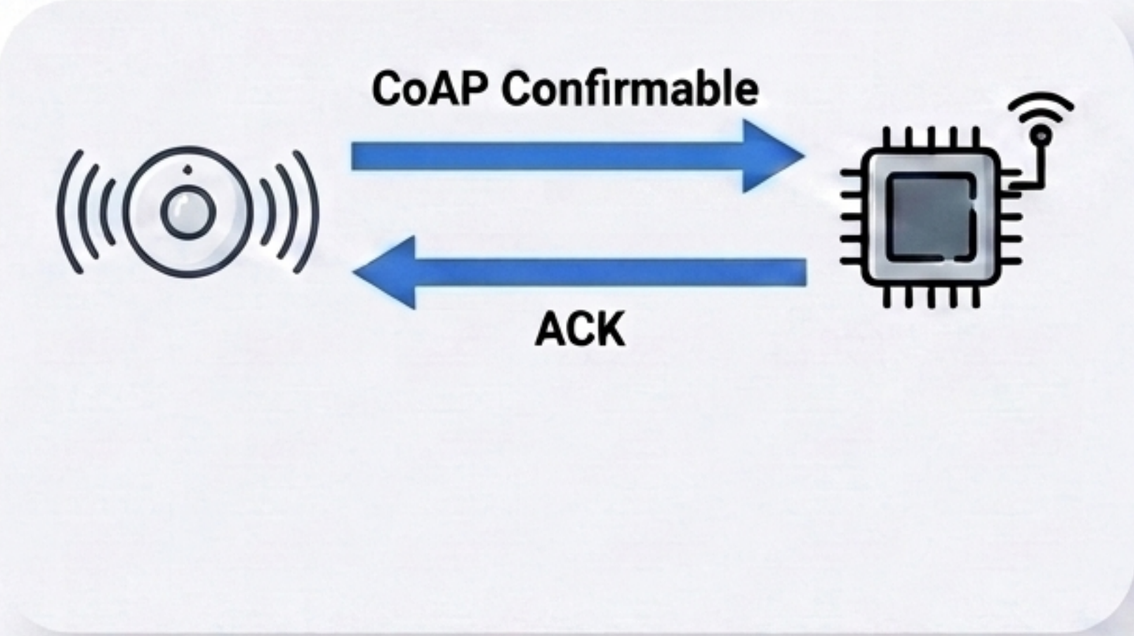
Mathematical Precision: Calculates Maximum and Standard Deviation locally using the ARM hardware Floating-Point Unit.

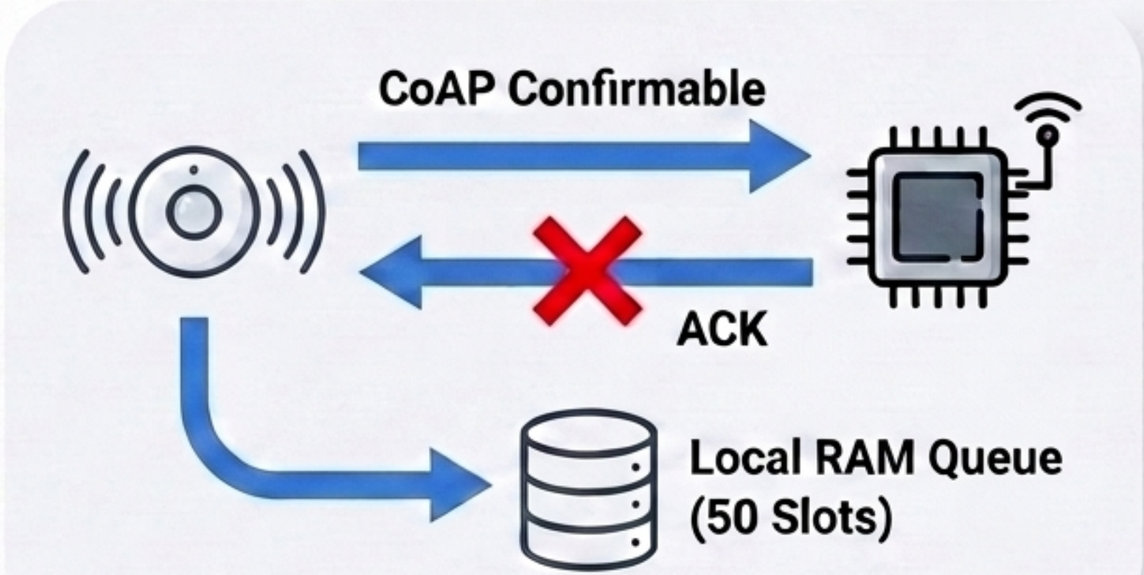
Event Awareness: Transforms raw numbers into definitive contextual events (e.g., rapid ventilation, active room cleaning) prior to transmission.





NotebookLM





The Mesh & The Safety Net

CoAP Confirmable:

Strict protocol demands a guaranteed handshake for every transmitted packet.

Autonomous Reflex:

3 consecutive failed CoAP handshakes trigger local memory queue buffering.

Temporary Local Data Preservation:

Microcontroller locally buffers packets safely for up to 25 minutes of temporary offline coverage, waiting to drain the queue upon reconnection.

NotebookLM

MULTI-LAYER AUTONOMOUS RESILIENCE

Sensor → Fog Failure Recovery

- Sensor generates CoAP packet
- Attempts packet transmission
- 3 consecutive handshake failures detected
- Firmware marks gateway unavailable
- Switches to local RAM queue
- Stores up to 50 packets locally
- Approximately 25 minutes autonomous coverage
- Fog connection restored
- Automatic queue draining begins
- Delayed packets successfully transmitted

Sensor firmware autonomously preserves packets locally whenever Fog Node connectivity becomes unavailable.

Fog → Backend Failure Recovery

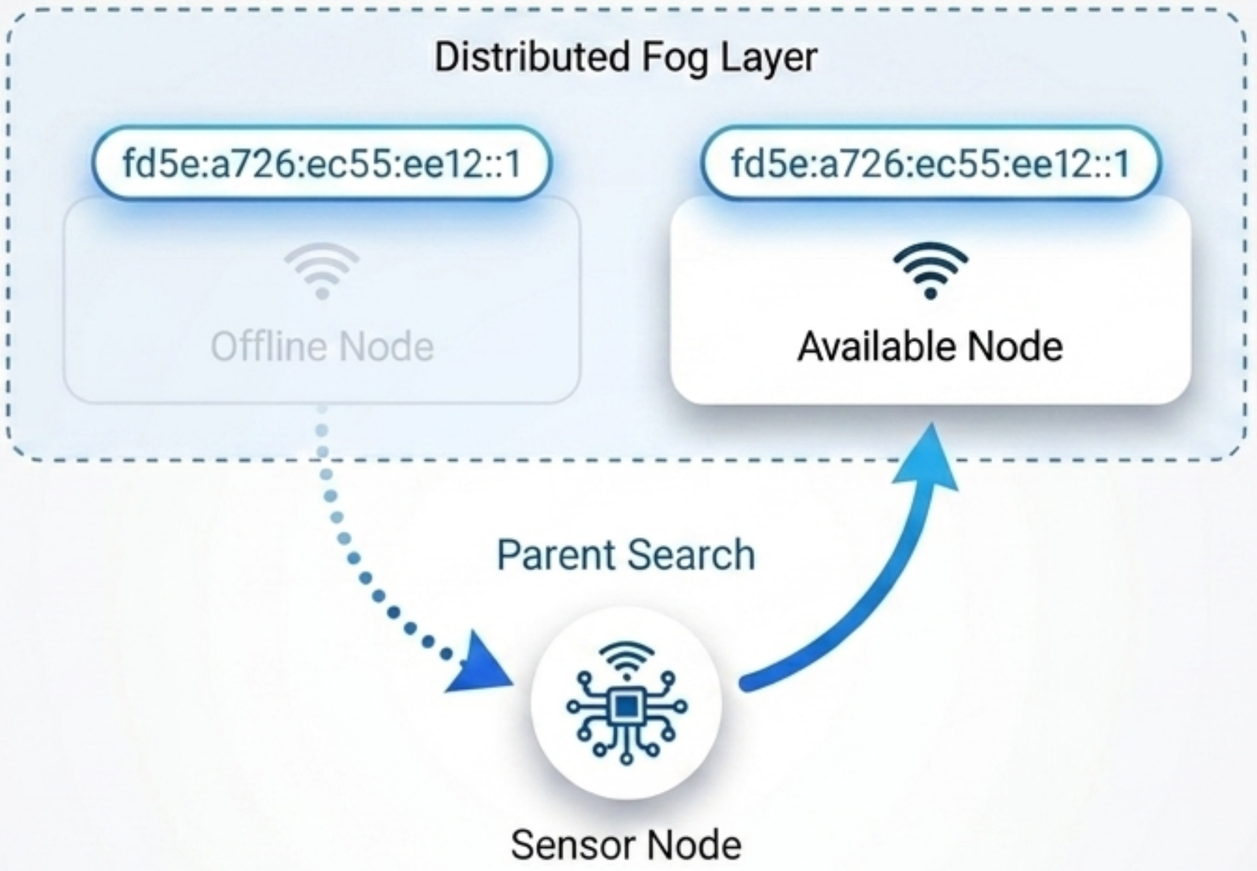
- Sensor packet successfully received by Fog Node
- Fog attempts backend forwarding
- Backend connection unavailable
- Packet stored in Redis cache
- Fog continues receiving sensor packets
- Wait for backend reconnection
- Backend connection restored
- Redis cache draining begins
- Delayed packets forwarded successfully

Fog Node autonomously preserves incoming packets whenever backend infrastructure becomes temporarily unavailable.

Two Independent Recovery Layers

Sensor-side RAM buffering protects against Fog Node unavailability. Fog-side Redis caching protects against backend connectivity failure. Together they provide end-to-end autonomous data preservation without manual intervention.

NotebookLM



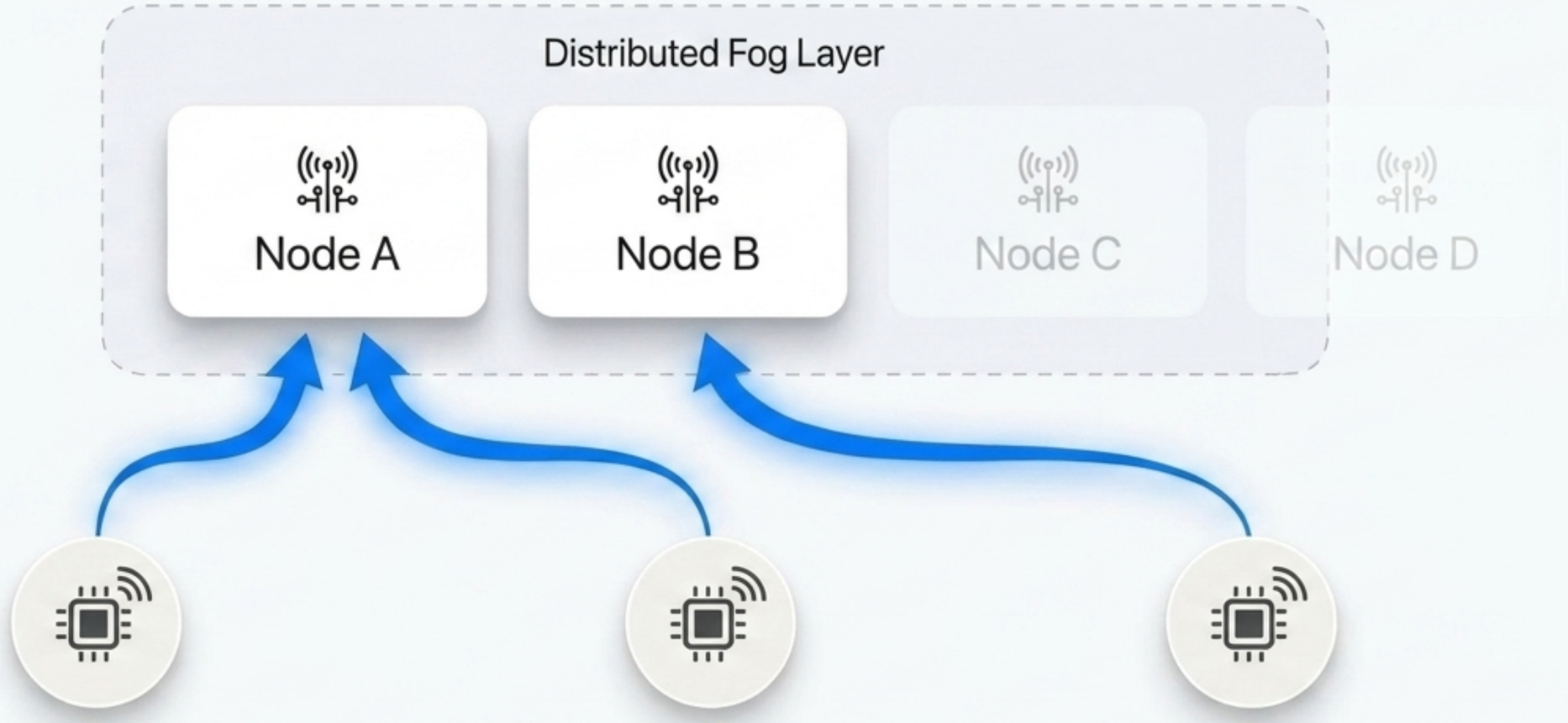
Shared Service Addressing

Intentional DAD Suppression: Multiple Border Routers are deliberately configured to share the exact same logical IPv6 service address.

Transparent Mesh Failover: The sensor executes an automatically initiates Parent Search and attaches to an available node in approximately 30 seconds.

Queue Drain: The 25-minute offline RAM buffer instantly uploads as soon as new connection is established.

NotebookLM



Scalable Distributed Fog Architecture

Dynamic Horizontal Scaling: New fog nodes can be provisioned endlessly without altering a single line of sensor code.

Distance-Vector Routing: The OpenThread mesh automatically calculates spatial distance, intrinsically routing packets to the physically nearest available node.

Zero-Configuration Traffic Distribution: Traffic balancing emerges organically from the physics of the mesh, enabling automatic multi-node traffic distribution.

NotebookLM

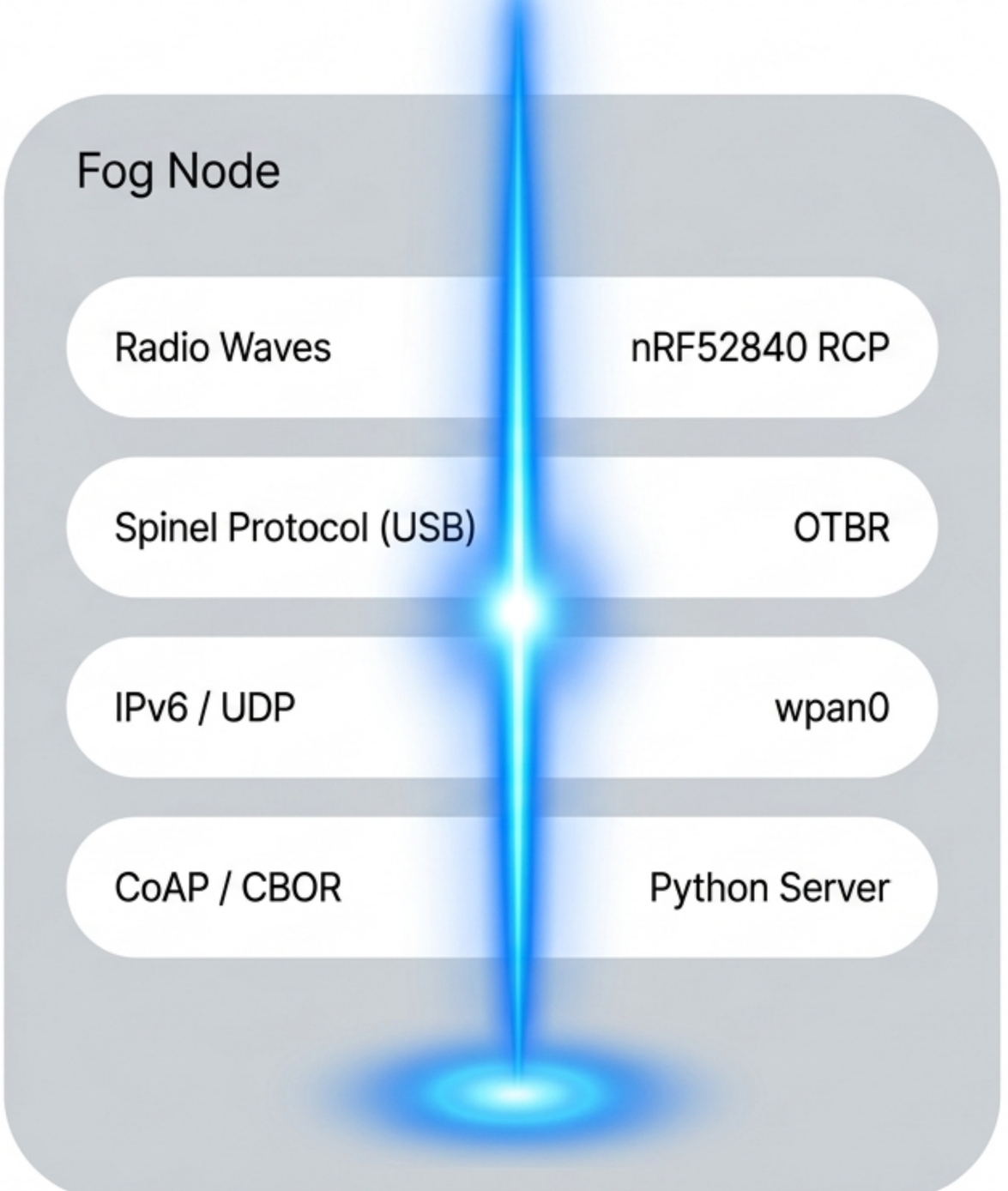
The Protocol Bridge

Radio Waves (nRF52840 RCP): The physical USB dongle acting as an antenna to catch raw 802.15.4 radio frames.

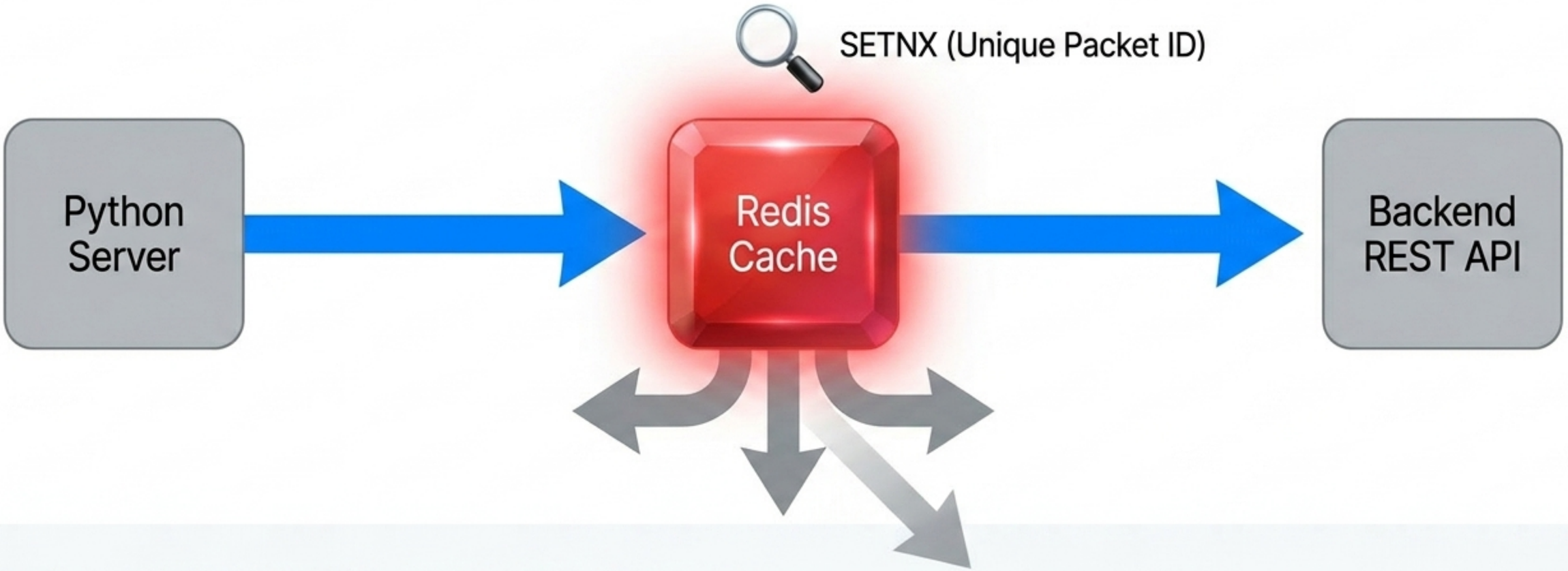
Spinel Protocol (OTBR): The OpenThread Border Router container unwrapping the serial data stream.

IPv6 / UDP (wpan0): The host operating system's virtual network interface routing the packet natively.

CoAP / CBOR (Python Server): The custom gateway software decoding the binary payload back into readable statistics.



NotebookLM

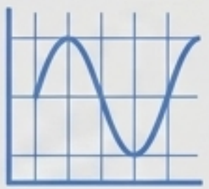


The Final Destination

- **Self-Contained Stacks:** Every fog node runs completely independent database queues to prevent cross-node contamination.
- **Atomic Deduplication:** Redis utilizes the SETNX command to instantly drop accidental double-transmissions from roaming sensors.
- **The Destination:** Clean, validated, and mathematically precise event data is queued for the enterprise REST API.

NotebookLM

Edge vs. Cloud: Architectural Justification



Data Continuity

Firmware maintains a perfect, unbroken mathematical window. Cloud-side statistical reconstruction becomes more complex when intermittent packet loss disrupts continuous sensor timelines.



Offline Resilience

Pre-calculated statistics allow sensors to bulk-upload delayed batches seamlessly after network outages
without requiring backend time-reconstruction.



True Edge Compute Distribution

Utilizing 64MHz ARM FPUs across thousands of edge nodes prevents a central server bottleneck when scaling to massive sensor swarms.

NotebookLM